

Qualitative segmentation and error correction across microscopy domains

Raw image

Ground truth

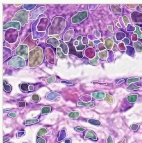
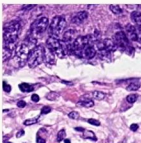
Cellpose cyto3

CellISAM

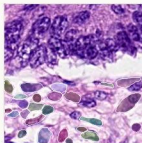
SAM-Cell

PanNuke

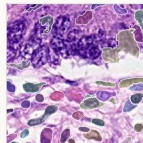
dense H&E nuclei
pannuke_fold3_1370



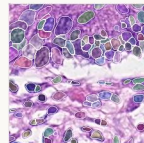
manual annotation



PQ 0.11 Dice 0.25



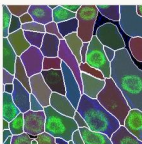
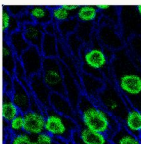
PQ 0.16 Dice 0.52



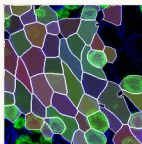
PQ 0.73 Dice 0.97

TissueNet

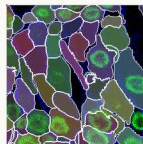
adherent tissue cells
tissuenet_val_279



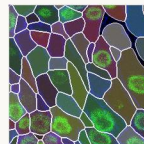
manual annotation



PQ 0.45 Dice 0.87



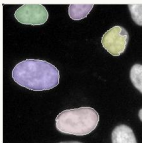
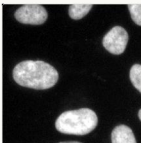
PQ 0.48 Dice 0.92



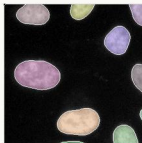
PQ 0.92 Dice 1.00

DSB2018

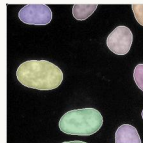
low-SNR fluorescence
dsb2018_adc315bd40d699f64e4



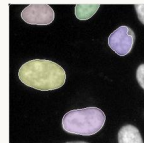
manual annotation



PQ 0.47 Dice 0.54



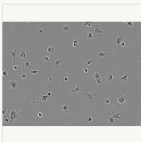
PQ 0.46 Dice 0.53



PQ 0.97 Dice 0.99

LIVECell

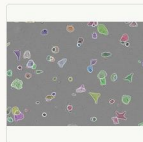
phase-contrast domain shift
livecell_A172_Phase_D7_2_00



manual annotation



PQ 0.67 Dice 0.90



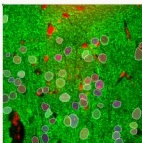
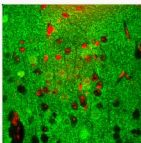
PQ 0.42 Dice 0.84



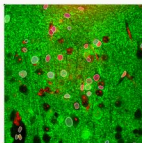
PQ 0.82 Dice 0.97

Cellpose

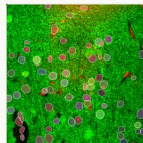
cell-culture morphology
cellpose_201



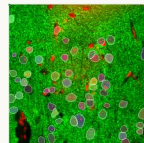
manual annotation



PQ 0.19 Dice 0.27



PQ 0.58 Dice 0.73



PQ 0.94 Dice 0.98